

# Predicting Per-Qubit Failure on Real IBM Chips with a Graph Neural Network

## Presentation Report

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### 1 What was Built

The quantum processor is modeled as a graph where nodes represent qubits and edges correspond to physical couplings executing two-qubit gates. A sample is defined as a **(device, qubit patch, circuit)** triple—representing the induced coupling subgraph of qubits onto which a specific circuit is compiled.

Node features fuse hardware characteristics with circuit demands:

- **Hardware metrics:**  $T_1$ ,  $T_2$ , readout error, single-qubit gate error, and incident two-qubit gate errors.
- **Circuit demands:** Gate counts, busy time, idle time, and schedule span.

### 2 Real IBM Hardware Data

Training data originates from recorded calibration snapshots of **54 real IBM Quantum processors** spanning Falcon, Hummingbird, Eagle, and Heron architectures (ranging from 5 to 156 qubits). All hardware metrics are derived directly from IBM's own physical measurements.

**Data Quality and Filtering:** Out of 67 available snapshots, 13 were dropped: one flagged as non-representative by Qiskit, one with failed two-qubit calibration, one single-qubit device, and ten older devices lacking recorded single-qubit gate errors. These were dropped rather than imputed to avoid fabricating training data.

- **Dataset Scale:** 7,020 samples · 41,192 labelled qubits · 12 circuit families · widths 4–9.
- **Label Distribution:** Median circuit fidelity is 0.966 (5th percentile 0.436, minimum 0.000). 43% of qubits exceed 1% infidelity, with the 99th percentile reaching 0.565.

### 3 Results and Performance

The Graph Neural Network (GNN) is evaluated against four distinct baselines: the training mean, a calibration error budget (analytic per-qubit estimate linearly recalibrated on the training set), gradient-boosted trees on identical features, and an MLP using identical features without message passing (serving as an ablation to isolate the utility of graph structure).

- **Ranking Performance:** The GNN wins clearly. Within-circuit Spearman correlation is **0.394**, compared to 0.308 for the feature-identical MLP and 0.116 for the calibration heuristic. It successfully identifies the single worst qubit in a circuit 44% of the time (versus a random guess rate of 11–25% for 4–9 qubit patches).

- **Absolute Magnitude:** The GNN does not win on absolute magnitude. Node MAE of 0.426 dex is statistically indistinguishable from the MLP’s 0.419 dex (run-to-run spread on repeated training is  $\pm 0.009$  dex). Message passing specifically optimizes the ordering of qubits within a circuit—the exact decision required by users.
- **Comparison to Current Practice:** The calibration budget is  $2.4\times$  worse in MAE and performs barely better than random at intra-circuit ranking ( $\rho = 0.116$ ). It identifies globally poor qubits, not those problematic for a specific circuit.

The in-distribution (random) split performs slightly better overall (MAE 0.388, within-circuit  $\rho = 0.410$ , P@m 0.638), indicating that performance transfer to unseen chips incurs minimal penalty.

## Application: Qubit Selection Efficiency

Tested on 10 held-out chips across 33 tasks (6 circuit families  $\times$  device) with 20 candidate qubit patches per task. Every candidate was simulated explicitly to determine true performance, while strategies made selections blindly.

| Strategy               | Mean True Fidelity | Mean Percentile in Pool | Picked Actual Best |
|------------------------|--------------------|-------------------------|--------------------|
| Random                 | 0.9476             | 60%                     | 6%                 |
| Calibration Budget     | 0.9627             | 83%                     | 27%                |
| <b>QChipGNN</b>        | <b>0.9637</b>      | <b>90%</b>              | <b>36%</b>         |
| Oracle (Best Possible) | 0.9692             | 100%                    | 100%               |

The GNN closes roughly 75% of the gap between random selection and the oracle (compared to 70% for the calibration heuristic). While absolute fidelity gains over the heuristic are modest (+0.001) because both avoid obvious failure patches, the GNN excels by pinpointing the absolute best patch more frequently (36% vs. 27%).

## 4 Scope and Limitations

**What this establishes:** Given real calibration data and a compiled circuit, a coupling-graph GNN ranks error-prone qubits substantially better than calibration tables alone or feature-equivalent models without message passing. It generalizes successfully to unseen processors, cross-generation architectures, and a wide scale range (5 to 156 qubits).

**What this does not establish:** Ground truth is derived from exact density-matrix simulation under IBM’s calibrated error model rather than physical hardware queue counts. Coherent/non-Markovian errors, spectator qubit crosstalk, and calibration-to-execution drift are excluded from labels, meaning these metrics represent upper bounds relative to physical hardware execution. Temporal drift is represented across diverse devices rather than longitudinally within a single device. Circuit widths are currently capped at 9 qubits due to simulation cost scaling, not architectural constraints.